Function: Formula Sheet

Domain

Sum, product: $D_f \cap D_g$

Quotient: $D_f \cap D_g - \{g = 0\}$

Even root: $f(x) \geq 0$

Log: $\log_g f$: $f > 0,\ g > 0,\ g \neq 1$

$\sin^{-1} u, \cos^{-1} u$: $-1 \leq u \leq 1$

${}^nC_r, {}^nP_r$: $n \in N,\ r \in W,\ r \leq n$

Composite:

$$D_{g \circ f} = \{x \in D_f : f(x) \in D_g\}$$

Standard functions

$\sin, \cos$: $R \to [-1, 1]$

$\tan$: $R - \{(2n+1)\frac{\pi}{2}\} \to R$

$\cot$: $R - \{n\pi\} \to R$

$\sec, \operatorname{cosec}$: range $|y| \geq 1$

Exponential: a^x: $R \to (0, \infty)$

Logarithm: $\log_a x$: $(0, \infty) \to R$

Modulus: $|x|$: $R \to [0, \infty)$

Signum: $\operatorname{sgn} x$: $R \to \{-1, 0, 1\}$

GIF: $[x]$: $R \to I$

Fractional part: $\{x\}$: $R \to [0, 1)$

$x + \frac{1}{x}$: $(-\infty, -2] \cup [2, \infty)$

GIF and fractional part

Bounds: $x - 1 < [x] \leq x$

Shift: $[x + m] = [x] + m,\ m \in I$

Sum: $[x] + [-x] = 0$ or -1

Sum: $\{x\} + \{-x\} = 0$ or 1

Inequality: $[x] + [y] \leq [x + y] \leq [x] + [y] + 1$

Equation: $[f] = k \iff k \leq f < k + 1$

Inequality: $[f] > k \iff f \geq k + 1$

Signum: $x \operatorname{sgn} x = |x|$

One-one, onto, counting

One-one: $f(a) = f(b) \Rightarrow a = b$

Cubic one-one: $b^2 \leq 3ac$

Onto: range = co-domain

Total functions: n^m

One-one ($m \leq n$): ${}^nP_m$

Onto ($m \geq n$):

$$\sum_{r=0}^{n} (-1)^r {}^nC_r (n - r)^m$$

$m = n$: one-one = onto = $n!$

Functional equations

Additive: $f(x + y) = f(x) + f(y) \Rightarrow f = kx$

Multiplicative: $f(x + y) = f(x)f(y) \Rightarrow f = a^x$

Log type:

$$f(xy) = f(x) + f(y) \Rightarrow f = k \log x$$

$f(x)f(\frac{1}{x}) = f(x) + f(\frac{1}{x})$: $f = 1 \pm x^n$

Pair sum: $f(x) + f(1 - x) = c$

Pair sum:

$$\sum_{k=1}^{n-1} f(\frac{k}{n}) = \frac{(n-1)c}{2}$$

Composite, even/odd, inverse

Composite: $(g \circ f)(x) = g(f(x))$

Even part: $\frac{f(x)+f(-x)}{2}$

Odd part: $\frac{f(x)-f(-x)}{2}$

Inverse of composite:

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

Identity: $f \circ f^{-1} = I$

Self-inverse $\frac{ax+b}{cx+d}$: $a + d = 0$

Linear/linear centre:

$$\left(-\frac{d}{c}, \frac{a}{c}\right)$$

Periods

$\sin, \cos, \sec, \operatorname{cosec}$: 2π

$\tan, \cot, |\sin|, |\cos|$: π

Odd powers $\sin^{2k+1} x$: 2π; even powers: π

$\{x\}$: 1; $|\sin x| + |\cos x|$: $\frac{\pi}{2}$

Linear argument: $T_{f(ax+b)} = \frac{T}{|a|}$

LCM of fractions: LCM of numerators $\div$ HCF of denominators